

Mapping the Narrow Corridor with Large Language Models

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Code: <https://github.com/esonghori/narrow-corridor-llm>

Gallery: <https://esonghori.github.io/narrow-corridor-llm/>

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Abstract

Acemoglu and Robinson (2024 Nobel laureates in Economics) trace national histories in *The Narrow Corridor* as paths through a two-dimensional (state power, society power) space but plot none: quantifying the two axes at each moment is the labor-intensive, judgment-laden task that has kept the framework qualitative. We ask whether large language models (LLMs) can operationalize it. We introduce a reproducible, provider-agnostic pipeline that scores a country period by period using chain-of-thought (events first, then score) and in-context anchoring on prior periods, against an explicit rubric with schema-validated output. As a proof of concept we produce a twelve-country trajectory atlas (Iran, France, the United Kingdom, the United States, China, Chile, Colombia, the Democratic Republic of the Congo, Lebanon, Zambia, Somalia, and India), chosen to place at least two countries in each of the book’s four Leviathan types, and compare four models. We assess the scores for consistency with an expert index (V-Dem) and for inter-model agreement, reading agreement as concurrent consistency rather than accuracy, since such indices likely appear in the models’ training data, and discuss conceptually how the biases of an LLM annotator and a human expert differ. Code, prompts, runs, and an interactive gallery of the animated trajectories are released.

1 Introduction

In *The Narrow Corridor* [2], Acemoglu and Robinson (awarded the 2024 Nobel Prize in Economics in part for this line of institutional research) propose that liberty is neither the natural product of a strong state nor of a strong society, but of a contested balance between the two. They picture a country’s history as a path through a plane whose horizontal axis is the power of civil society and whose vertical axis is the power of the state; a “narrow corridor” running between the axes

marks the region where the two powers grow together and liberty is sustained. The book is rich in narrative but contains *no plotted trajectories*. The obstacle is not conceptual but operational: turning centuries of a nation’s history into a sequence of (society, state) coordinates requires an enormous amount of expert judgment, and any such coding is vulnerable to the biases, selective readings, and coarse simplifications that historians rightly distrust.

This is a measurement problem, and measurement problems in the social sciences increasingly admit machine assistance. Large language models have been shown to annotate and classify political text at or above the level of crowd workers [11], and a growing literature examines their use as instruments in computational social science [19]. We ask a narrower, concrete question: *can an LLM place a country in the Narrow Corridor space, period by period, in a way that is reproducible, checkable against an established expert index, and useful to a researcher?*

We take an applied, digital-humanities stance. Our primary artifact is an **twelve-country trajectory atlas** (Iran, France, the United Kingdom, the United States, China, Chile, Colombia, the Democratic Republic of the Congo, Lebanon, Zambia, Somalia, and India), rendered as time-shaded paths through the corridor, together with the cross-country patterns those paths reveal. The twelve are chosen so that at least two countries fall in each of the book’s four Leviathan types (Section 4), which lets us ask whether an LLM places each country in the region the book assigns it. The method and its validation are the backing that makes the atlas trustworthy rather than decorative. Concretely, we contribute:

1. **A scoring method.** A period-by-period pipeline combining an anchored 0–10 rubric, chain-of-thought elicitation of events and trends [18], and in-context anchoring on prior periods [5, 8].
2. **A provider-agnostic, open implementation.**

Any model reachable through a unified interface can be swapped by changing one string; we release code, prompts, caches, and every run.

3. **A trajectory atlas and cross-country reading** of twelve countries spanning the book’s four Leviathan types.
4. **An interactive, publicly hosted web gallery.** A static site (GitHub Pages) presenting every trajectory as a time-shaded plot with *click-to-play animations* that reveal the path period by period and annotate the historical event driving each move; readers explore and compare models directly in the browser, turning a static figure into a navigable atlas.
5. **An assessment of the scores** for consistency with an expert index (V-Dem) and for inter-model reliability, with explicit treatment of the training-data-overlap threat that makes such agreement concurrent consistency rather than proof of accuracy.
6. **A conceptual discussion of measurement bias:** what an LLM annotator plausibly gets right and wrong relative to a human researcher or an expert panel doing the identical task. We do not commission a fresh panel of coders; V-Dem is itself exactly such a study, a large expert coding operation with many raters per case, so our comparison against it (Section 5.4) already stands in for the human-expert baseline.

2 Related Work

The Narrow Corridor and measuring state/society power. The (state, society) framing extends the institutional program of Acemoglu and Robinson [1, 2]. Quantifying the two axes connects to decades of effort to measure regime characteristics and state capacity: the Varieties of Democracy (V-Dem) project aggregates expert codings into latent indices including a core civil-society index and measures of state administrative capacity [6, 16]; Polity5 codes regime authority on an autocracy–democracy scale [14]; and Freedom House rates political rights and civil liberties [10]. These projects are themselves large expert coding operations, which makes them both our natural validation targets (Section 5.4) and our natural comparison point when we ask what an LLM changes about the coding process (Section 6).

Text as data and LLMs as social-science instruments. The use of automated methods to extract measures from text is well established [12]. Recent work reports that instruction-tuned LLMs match or exceed

crowd annotators on political text-labeling tasks [11] and surveys their broader promise and pitfalls for computational social science [19]. A related strand uses LLMs to *simulate* human populations [3] and debates whether they can stand in for human participants at all [7]. Our task differs from labeling or simulation: we ask the model to produce a continuous, temporally coherent *measurement* of a theoretical construct, then check that measurement against independent data.

Prompting. Our pipeline relies on two now-standard techniques. Chain-of-thought prompting [18] improves reasoning by eliciting intermediate steps; we use it to force the model to enumerate historical events and trends *before* scoring. In-context learning [5, 8] lets the model condition on examples provided at inference time; we feed all prior periods’ scores forward so the trajectory is temporally coherent rather than a sequence of independent guesses. This anchoring is a deliberate design choice with a cost (an early mis-score can propagate) that we return to in Section 7.

Bias and reliability. LLMs inherit biases from training data [4] and exhibit measurable political leanings: they do not reflect a representative cross-section of opinion [17], can carry the political slant of their pretraining corpora into downstream behavior [9], and have been probed directly for partisan bias [15]. These findings motivate both our reliability analysis, which borrows the logic of inter-coder reliability from content analysis [13], and our discussion of how LLM bias compares to the well-known biases of human expert coders.

3 Method

The space. We place a country at each time period t at a point $(soc_t, sta_t) \in [0, 10]^2$, where soc_t is the power of civil society and sta_t is the power of the state. A period is a fixed span of years (e.g., a decade). We read a country as *in the corridor* in period t when its state and society power are close (near the diagonal) and *above* or *below* it when one clearly dominates. Following the book, the corridor in our plots is *widening*: the dashed boundaries are narrow where state and society are both weak (bottom-left) and diverge as the two powers grow together (top-right), because liberty becomes easier to sustain once both are strong. They mark the band illustratively, not as a fitted threshold, so positions are read qualitatively, from where a path sits relative to the widening band, not from a numeric cutoff. Figure 1 sketches the space and the book’s four Leviathan regions in this style.

Anchored rubric. A free-floating request for “two numbers” is not a measurement. Every scoring prompt

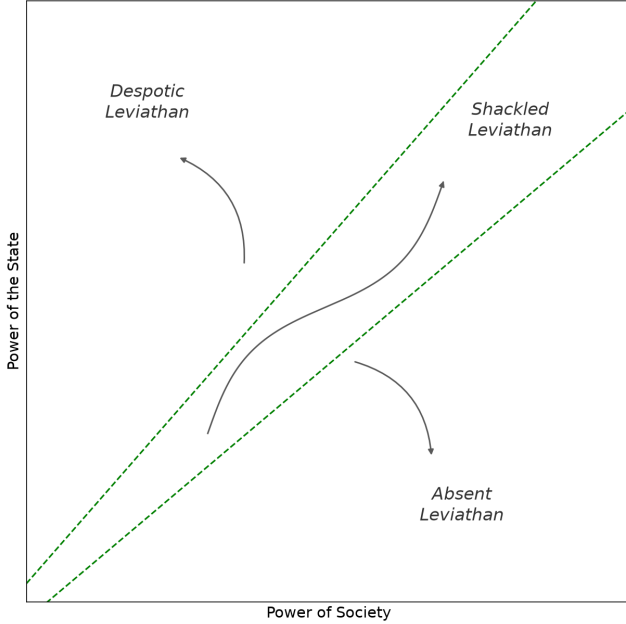


Figure 1: The Narrow Corridor [2]. The dashed lines are the book’s *widening* corridor; a country whose path stays inside it is a Shackled Leviathan, one above it a Despotic Leviathan, and one in the weak-state region below it an Absent Leviathan.

embeds a fixed 0–10 rubric that defines each band of state power (from 0–2, collapsed/nonexistent, to 9–10, overwhelming penetration of society) and society power (from 0–2, atomized, to 9–10, society pervasively checks the state), plus explicit neutrality guidance: judge each period on its own terms, weigh evidence for and against a shift, avoid *presentism* (scoring a period by how things later turned out, or by today’s standards, rather than by the conditions and information of the period itself), and treat the two axes as independent. The rubric is what makes scores comparable across periods, countries, and models.

Chain-of-thought: events then scores. For each period the model is first asked to list the major events and trends affecting state or society power, separating slow-moving trends from discrete events and noting the direction of each effect. This narrative is then inserted into a second prompt that requests the numeric score, so the number is conditioned on explicit reasoning rather than produced in one leap [18].

In-context learning: temporal anchoring. Starting from the earliest period (scored on absolute terms), we accumulate every prior period’s (soc, sta) pair and feed the running trajectory into each subsequent scoring prompt [5]. The model reports both the change from the previous period and the resulting absolute val-

ues, which discourages the sequence from drifting and keeps a quiet period near zero change.

Structured, validated output. Each scoring call returns a JSON object (the two power values, their per-period changes, a short key event, and a one-sentence justification) validated against a fixed schema. A malformed response triggers one reminder retry and then a numeric fallback extractor before an error is raised.

Provider-agnostic access. All model calls go through a single unified interface, so a model is selected by one identifier string and providers can be swapped without code changes. This is what lets us run the identical prompts across four model families and isolate model effects from prompt effects.

Visualization. A run renders as (i) a static plot of society vs. state power with the path time-shaded from early (dark) to late (bright) and a year colorbar, the largest moves annotated with their year and key event; and (ii) an animation that reveals the path period by period, drawing an arrow for the move each event caused and captioning it. Figure 2 is built from the static renderings; the animations are best viewed in the interactive gallery at <https://esonghori.github.io/narrow-corridor-llm/>.

4 Experimental Setup

Countries. We study twelve countries, chosen so that at least two fall in each of the four Leviathan types the book defines. **China** and **Iran (Persia)** are candidate *Despotic* Leviathans (a strong state over a weak society); the **United Kingdom**, **France**, the **United States**, and **India** are candidate *Shackled* Leviathans (state and society balanced inside the corridor, India’s under the “cage of norms” the book stresses); **Colombia**, **Zambia (Northern Rhodesia)**, and **Lebanon** are candidate *Paper* Leviathans (a state that exists on paper, a constitution and formal institutions, but cannot enforce its authority); and the **Democratic Republic of the Congo** and **Somalia** are candidate *Absent* Leviathans (a state that is effectively absent, lacking even the forms of one). **Chile** is included as a leave-and-re-enter case around the 1973 coup. Crucially, the Paper and Absent types are *not* distinguished by position in the (society, state) plane, both are weak states low on the state axis, but *de jure*: a Paper Leviathan has the forms of a state it cannot enforce (Lebanon), an Absent Leviathan lacks even those (Somalia). Our figures therefore label only the three spatially distinct regions (Despotic above the corridor, Shackled inside it, and the weak-state region below); the Paper/Absent distinction is drawn here in the text. Labeling each a

“candidate” is deliberate: whether the LLM actually places a country in its assigned region, and in the corridor’s expected part, is a question we test, not an assumption (Section 5.1). Each country is scored from a country-appropriate start year through 2020 (a common cutoff for consistency across cases) in decade (10-year) periods, giving 11–24 periods per country (Table 1 lists windows and counts). Twelve countries is a proof of concept, not a basis for cross-national generalization.

Models. We compare four models from four families (Google, Anthropic, OpenAI, Alibaba): three closed-weight models (*gemini-2.5-pro*, *claude-opus-4-8*, *gpt-5.5*) and one open-weight model (*qwen-2.5-72b-instruct*), which keeps an open- vs. closed-weight contrast. All are driven through the same interface with identical prompts, isolating model effects from prompt effects.

Consistency check against an expert index. We compare the LLM scores to V-Dem [6], aligning each period’s midpoint year to the V-Dem year. We map LLM *society power* to the V-Dem core civil-society index (*v2xcs_ccsi*) and LLM *state power* to a state-authority measure (V-Dem state authority over territory, *v2svstterr*), a narrow, territorial-control proxy for the rubric’s broader penetration-of-society notion of state power. The mapping is fixed before inspecting correlations, and alternative capacity indicators are easy to swap in the released code. We report the *Spearman rank correlation* ρ per country, a number from -1 to $+1$ that measures whether two series move up and down together in the same rank order ($+1$ perfect agreement, 0 none, -1 opposite), computed on both *levels* and *first differences* (period-to-period change): levels are dominated by a shared secular trend and inflate the correlation, so first differences are the more honest agreement signal. Crucially, V-Dem is itself almost certainly in the models’ training data, so we read this as *concurrent consistency with expert consensus*, not as external proof of accuracy (Section 7). Polity5 and Freedom House are natural additional anchors but are out of scope here.

5 Results

5.1 Trajectory atlas

Figure 2 shows the twelve trajectories under *claude-opus-4-8*, chosen as a single reference model for legibility; every model’s per-country panels are in the online gallery, with cross-model disagreement quantified in Section 5.5. Read against the book’s four Leviathan types, most placements land where the framework expects. The **United Kingdom**

stays inside the corridor throughout, state and society rising together (the Red Queen), a textbook Shackled Leviathan. **France** enters it, moving from society-leaning in 1789 to inside by the late twentieth century, Shackled by its late history. **Iran** and **China** exit *upward* into state dominance, Iran after 1979 and China after 1949, the Despotic Leviathan the book assigns them. **Chile** loops out and back, a sharp fall of society and spike of state at the 1973 coup, then a return toward the corridor after the 1990 transition. **Colombia**, the book’s Paper Leviathan, rises from a weak post-independence founding to only moderate levels by 2019 (society ≈ 6.4 , state ≈ 5.7 , the 1886 centralist constitution its largest move), never consolidating the strong, high-capacity state of the Shackled cases; it ends in the lower corridor rather than climbing to the strong-both corner, the muted arc the Paper-Leviathan reading predicts, though the model treats late-twentieth-century Colombia as closer to the corridor than clearly below it. The **Democratic Republic of the Congo**, the book’s Absent Leviathan, stays near the origin throughout: state power spikes briefly under Mobutu’s mid-century regime, then collapses around 1960 independence and the subsequent wars, ending low on both axes (society ≈ 3.2 , state ≈ 3.2), the persistently weak state the Absent-Leviathan label describes. The **United States** is another Shackled Leviathan, much like the UK: state and society grow strong together and it ends inside the corridor, if with a somewhat stronger state tilt than the UK. Its state axis climbs faster than the UK’s, but it stays within the corridor rather than crossing into state dominance. **Lebanon** and **Somalia** end deep in the weak-state region below the corridor (society ≈ 4.8 and 4.6 , state ≈ 1.9 and 2.6), a society outrunning a state that is barely present: Lebanon’s state erodes across the 1975–90 civil war and the post-2019 collapse, Somalia’s stays low and flat throughout. They illustrate the Paper/Absent point directly, sitting in the same low-state region while differing only *de jure* (Section 4). **Zambia** is the case that may surprise readers unfamiliar with its recent history: hypothesized a Paper Leviathan, the models instead carry it *into* the corridor by 2020 (society ≈ 5.8 , state ≈ 6.0), reading its post-1991 multiparty democracy and its peaceful transfers of power as genuine balanced growth rather than a hollow paper state, a reading that is defensible on that history even though it does not match the label we assigned in advance. **India** lands where the book places it, a Shackled Leviathan (society ≈ 8.0 , state ≈ 8.5): a strong colonial state in 1885 that its society gradually catches and checks, ending high on both axes. Long diagonal runs mark co-growth; tight loops mark oscillation in contested periods; large single-period jumps mark ruptures.

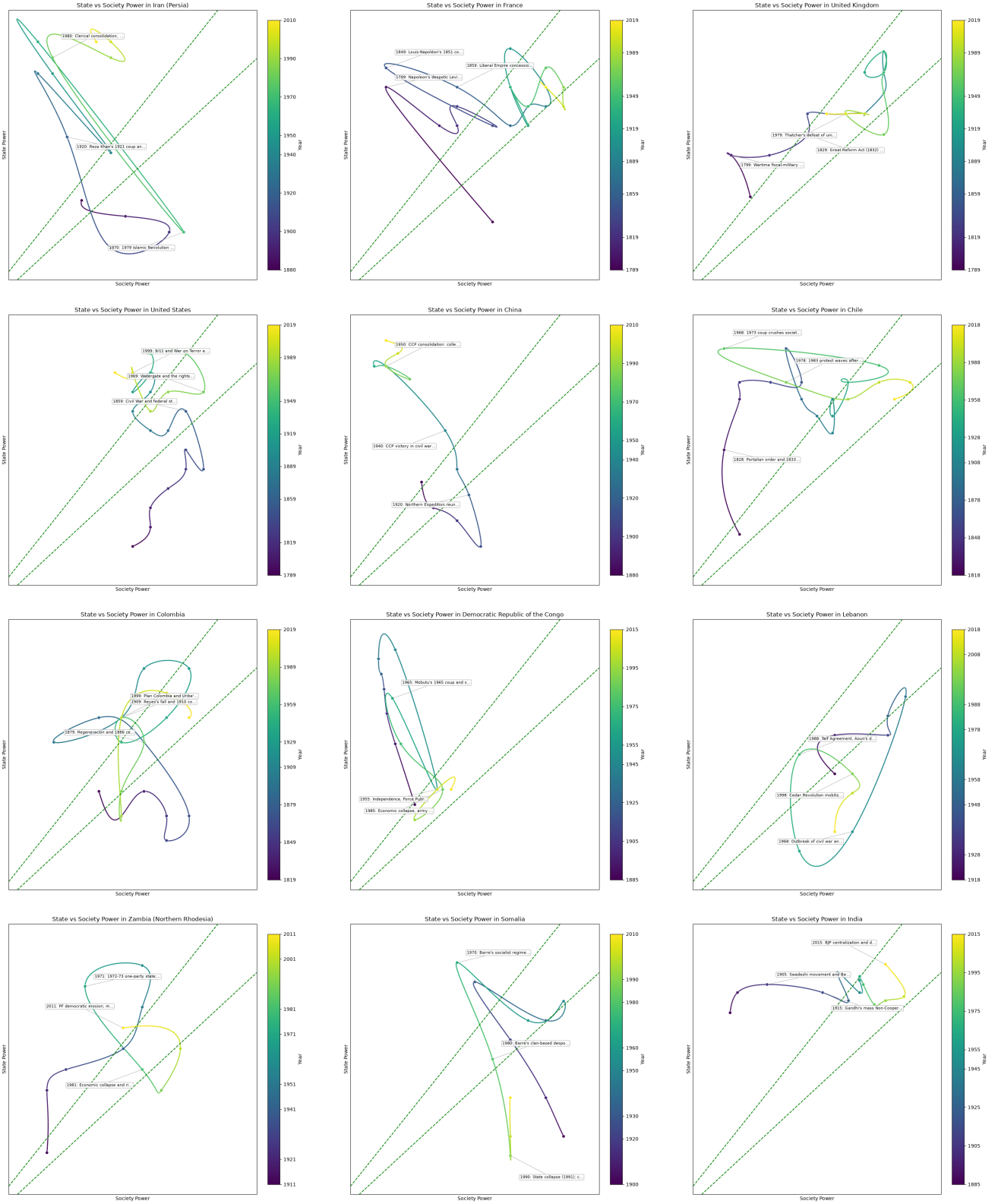


Figure 2: Trajectory atlas (reference model **claud-opus-4-8**): society power (x) vs. state power (y) for all twelve countries, time-shaded (dark = early, bright = late); dashed lines mark the book's widening corridor illustratively. Animated versions are in the online gallery.

5.2 Ensemble atlas (mean across models)

Figure 3 shows, for each country, the trajectory averaged across all four models period by period. Because independent models rarely share the same idiosyncratic error, the mean is a more robust reading than any single model’s path: where the four agree it tracks the consensus tightly, and where they diverge it sits between them, so a point should then be read as uncertain rather than precise (the underlying per-axis agreement is quantified as Krippendorff’s α in Section 5.5). Each panel is titled “(ensemble mean)” to distinguish it from the single-model atlas of Figure 2. We treat this mean atlas, not any single model, as the paper’s headline reading. The mean is close to the reference model for the high-agreement cases (Iran, Chile; Section 5.5) and visibly smoother for the United States, where the models disagree most about how fast its state grew. The corridor placements are unchanged: the UK, France, and the US inside the corridor, Chile entering it, Iran and China state-dominated, and Colombia and the Congo low.

5.3 One map, all countries

The book plots countries together in a single plane. Doing the same requires the scores to be comparable *across* countries, not merely across periods within one country, and this is a real question rather than a given: does the shared rubric actually put every country on one scale? Our design bets that it does. The fixed 0–10 anchors (Section 3) are country-independent, so a 7 on the state axis is meant to denote the same level of state capacity for the Congo as for China. Figure 4 overlays all twelve ensemble-mean paths in one space to test that bet. The relative placement is broadly sensible: the Despotic cases (China, Iran) sit high on the state axis with weak society, the Shackled cases (UK, France, the US, and India) climb the diagonal into the strong-both upper-right, and the weak-state cases fall below and left of the corridor, the Congo, Somalia, and Lebanon down in the low-state region (society ahead of a barely-present state) and Colombia at the corridor’s lower edge. The one country the models place outside its candidate region is Zambia, inside the corridor rather than below it as a Paper Leviathan, a reading that fits its stable post-1991 democracy. That the countries otherwise sort into the book’s regions on a single common scale is evidence the rubric transports across countries well enough for a joint reading. It is not proof: the shared training-data biases that may inflate a score could do so consistently across countries (Section 7), and the absolute levels are softer than the within-country ordering. We therefore read the joint map as a qualitative cross-country atlas, not a metric one.

Three regularities recur. First, the countries the book treats as corridor-dwellers (the UK throughout, France and Chile by their late histories) trace long diagonal runs in which state and society grow together. Second, every country’s largest single-period move is an institutional rupture that pushes the *state* axis up: the 1979 Iranian revolution, the 1973 Chilean coup, post-1949 Chinese consolidation, and the US Civil War. Third, the two twentieth-century revolutions (Iran 1979, China 1949) end in state dominance rather than liberty. The mechanism the paths trace is not that mobilizing society produces despotism, which would invert the book’s logic, but the opposite: a revolution briefly raises society’s power, then the new revolutionary state consolidates and *demobilizes* the society that carried it, outgrowing rather than balancing it. Liberty fails because society does not stay strong enough to shackle the state the revolution builds, which is the book’s Despotic Leviathan path.

5.4 Consistency with V-Dem

Table 1 reports correlations between the LLM scores and V-Dem over overlapping years, on both levels and first differences. On *levels*, LLM society power tracks V-Dem’s core civil-society index strongly for China ($\rho = 0.89$), France (0.87), Chile (0.74) and the UK (0.70); the one low value, the United States ($\rho = -0.10$), is a variance artifact rather than a category mismatch, since V-Dem holds US civil society nearly flat and high across the whole window, leaving almost nothing for a rank correlation to track. On the more demanding *first differences* the society signal is positive but weaker (China 0.66, Iran 0.60, France 0.40), as expected once the shared trend is removed. The *state* axis is where consistency collapses: first-difference $\rho \approx 0$ for most countries. Figure 5 shows why: our V-Dem state proxy (`v2svstterr`, authority over territory) is near-saturated and flat for consolidated states, so it cannot move with the LLM’s broader state-*capacity* notion; the LLM tracks the *building* of the modern state that a territorial-control index does not. As noted above, this measures agreement with expert consensus the models were likely trained on, not independent accuracy.

Figure 5 re-plots each country from V-Dem in the same space (society = `v2xcs_ccsi`×10, state = `v2svstterr`/10), at the LLM period-midpoint years. The society axis broadly rhymes with the LLM ensemble (Fig. 3), but the state axis is pinned high for every consolidated state: V-Dem places even 1880s Qajar Iran and the early United States near the top of the state axis, the opposite of the LLM, which reads those as weak states still building capacity. The comparison is thus best read as *agreement on society, divergence on the operationalization of state power*.

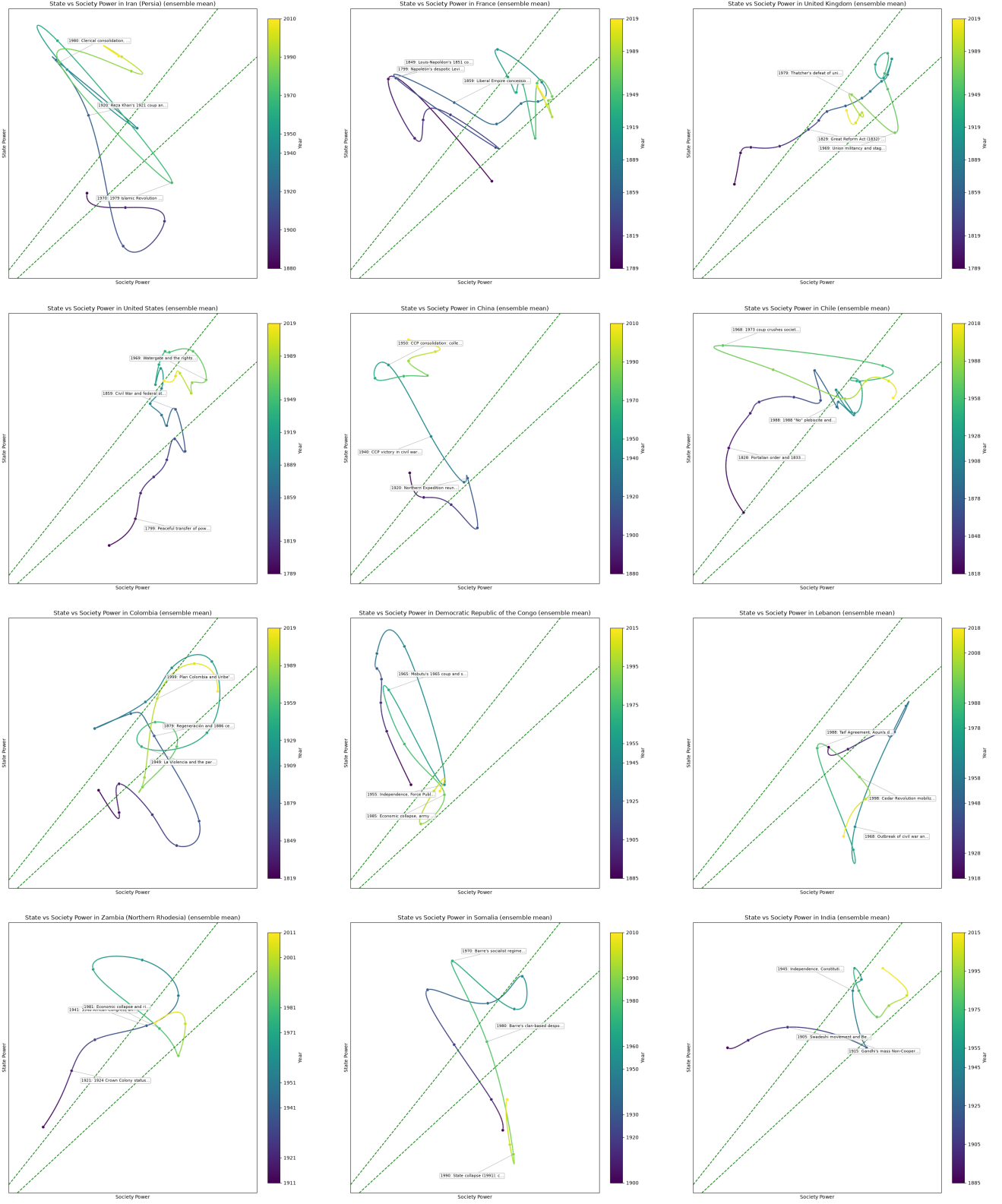


Figure 3: Ensemble atlas: the same twelve countries, each averaged across the four models period by period (panels titled “ensemble mean”). The mean damps any one model’s idiosyncrasy; the spread it averages over is Krippendorff’s α in Table 2. Time-shaded; dashed lines mark the widening corridor.

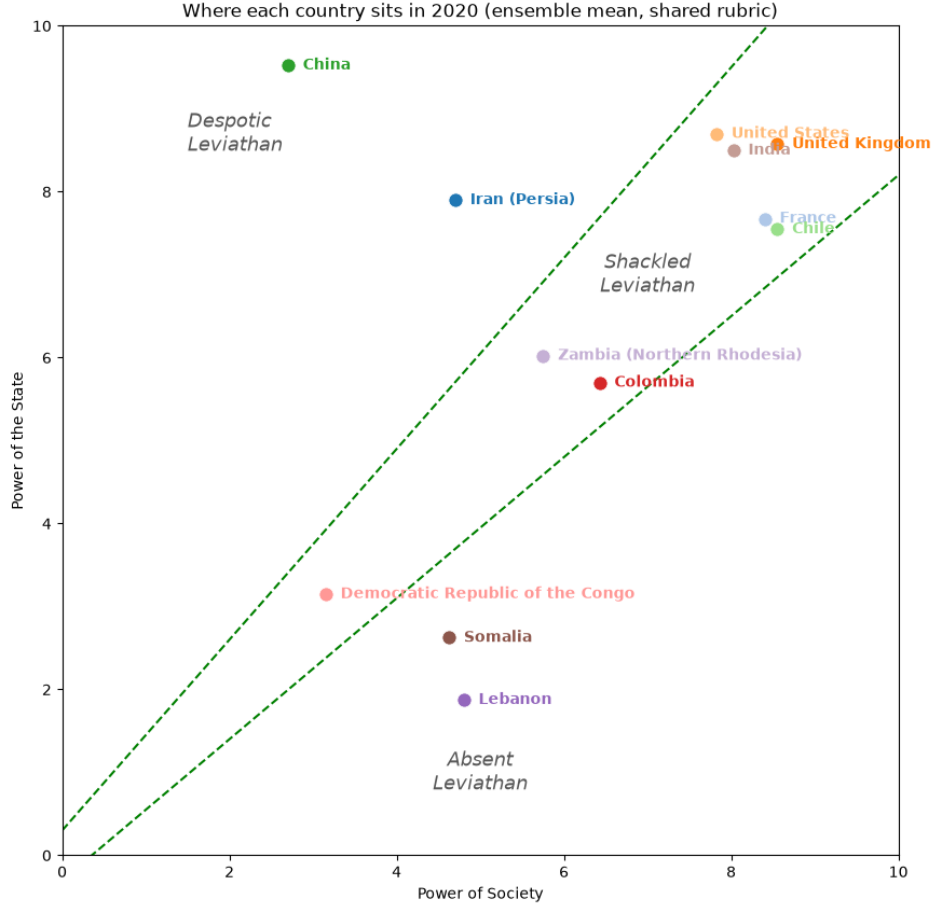


Figure 4: All twelve countries’ ensemble-mean trajectories in one shared (society, state) space, the direct analogue of the book’s single-plane map. An open marker is each country’s start year, a filled marker and label its last period; dashed lines mark the book’s widening corridor. Because every country is scored against the same fixed rubric, the paths are plotted on a common scale without any per-country rescaling.

5.5 Model comparison and reliability

Treating each model as a “rater” in the sense of inter-coder reliability [13], Table 2 reports agreement between models on the same country. We report *Krippendorff’s* α , a standard inter-coder reliability coefficient that measures how much raters (here, the four models) agree beyond what chance would give: $\alpha = 1$ is perfect agreement, 0 is chance, and negative values mean systematic disagreement, with $\alpha \geq 0.8$ conventionally read as strong and 0.67–0.8 as tentative. We compute it (interval variant, treating the 0–10 scores as interval-scaled) on the period-to-period *changes* rather than on levels (for the same reason first differences are used in Section 5.4): agreement on changes tests whether models see the same *moves*, not merely the same secular trend. Models agree slightly more on society ($\alpha = 0.70$ overall) than on the state axis (0.68), and most on the countries with the clearest arcs, Iran (0.77/0.78) and Chile (society 0.83). Agreement is lowest for the United States (society $\alpha = 0.47$), but the disagreement is one of de-

gree, not of category: all four models place it inside the corridor as a Shackled Leviathan and differ only on how strong its central and security state grew. The open-weight model does not systematically diverge on the periods it scored.

5.6 Qualitative case studies

Iran, 1979. All four models place a society peak in the 1970s that collapses into state dominance in the 1980s (the ensemble society score falls sharply while state climbs past 8). This matches the book’s account of the revolution: the mobilization that toppled the Shah briefly raised society’s power, but the new revolutionary state then consolidated and demobilized that society rather than balancing it, ending in a Despotic Leviathan. **Chile, 1973/1990.** The 1973 coup is the largest single move (society crashing, state spiking near the ceiling), followed by recovery after the 1990 plebiscite, a country leaving and re-entering the corri-

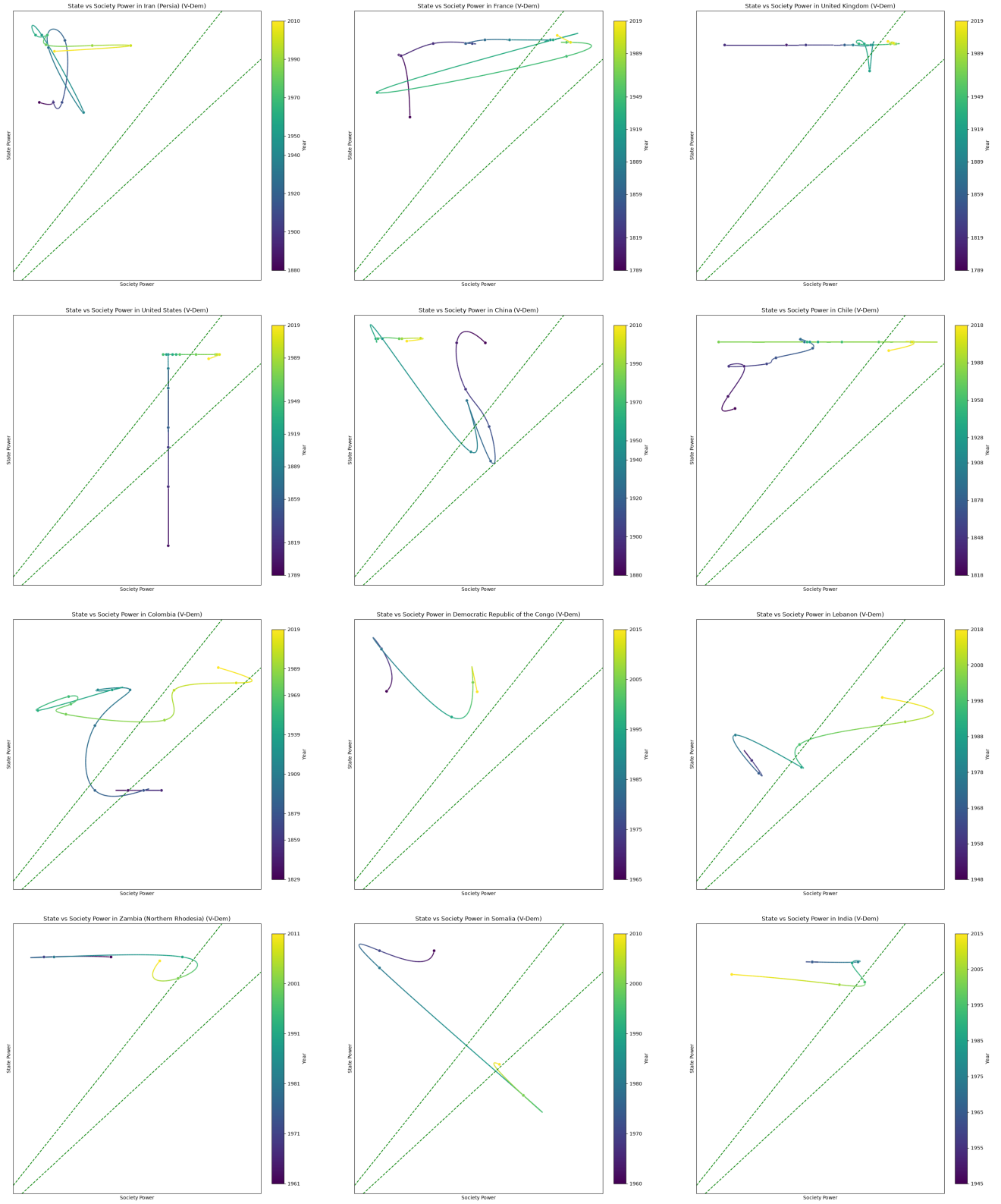


Figure 5: V-Dem atlas: the same twelve countries from expert codings (society = core civil-society index $\times 10$; state = authority over territory /10) at the LLM period-midpoint years. The society axis broadly agrees with the LLM ensemble (Fig. 3), but V-Dem’s territorial-authority state axis is near-saturated for consolidated states and misses the state-capacity dynamics the LLM captures.

Table 1: Consistency with V-Dem (claude-opus-4-8): Spearman ρ at period-midpoint years, reported as level/ Δ (levels vs. first differences). Society vs. `v2xcs_ccsi`; state vs. `v2svstterr`. First differences are the more honest signal.

Country	Window	N	Society ρ	State ρ
Iran (Persia)	1880–2020	14	0.45 / 0.60	0.42 / 0.13
France	1789–2020	24	0.87 / 0.40	0.20 / 0.04
United Kingdom	1789–2020	24	0.70 / 0.21	0.13 / 0.07
United States	1789–2020	24	-0.10 / 0.26	0.84 / 0.15
China	1880–2020	14	0.89 / 0.66	0.74 / 0.24
Chile	1818–2020	21	0.74 / 0.17	0.18 / 0.42
Colombia	1819–2020	20	0.46 / 0.51	0.79 / 0.41
DR Congo	1885–2020	6	0.46 / -0.30	0.52 / -0.21
Lebanon	1918–2020	8	-0.20 / 0.24	-0.39 / -0.33
Zambia	1911–2020	6	0.75 / 0.79	0.03 / -0.23
Somalia	1900–2020	6	0.56 / 0.76	0.97 / 0.79
India	1885–2020	8	0.31 / 0.05	0.41 / 0.54

Table 2: Inter-model reliability: Krippendorff’s α (interval) on period-to-period *changes* across models, per country. Higher = models see the same moves.

Country	#models	Society α	State α
Iran (Persia)	4	0.77	0.78
France	4	0.72	0.56
United Kingdom	4	0.56	0.63
United States	4	0.47	0.62
China	4	0.56	0.56
Chile	4	0.83	0.57
Colombia	4	0.54	0.53
DR Congo	4	0.51	0.77
Lebanon	4	0.34	0.71
Zambia	4	0.61	0.82
Somalia	4	0.56	0.83
India	4	0.47	0.16
Overall	–	0.66	0.68

dor. **United States.** Like the UK, the US traces a long co-growth of state and society, ending inside the corridor as a Shackled Leviathan, with the Civil War its largest single move. It sits a little higher on the state axis than the UK, reflecting the twentieth-century build-out of the federal and security state, but it does not cross into state dominance. The models agree least here (Section 5.5), differing on how strong that central state grew relative to civil society, yet they concur on the corridor placement, so the US reads as a second Shackled case alongside the UK rather than an exception.

6 Discussion: LLM vs. Human Researcher or Expert Panel

Both an LLM and a human expert (or panel) performing this coding are biased; the useful question is *how the bias profiles differ*, and which task properties favor which coder.

What a human panel brings. Domain expertise, sensitivity to contested historiography, the ability to flag “this case is genuinely disputed,” and accountable, attributable judgment. Established indices such as V-Dem manage individual coder bias with multiple experts per case and a latent measurement model that estimates and corrects for coder disagreement [16]; content analysis formalizes this as inter-coder reliability [13]. The costs are equally real: expert panels are slow, expensive, hard to scale to arbitrary country-periods, and carry their own systematic biases (national, ideological, disciplinary) that are difficult to audit because they live in individual judgment.

What an LLM changes. An LLM annotator is fast, cheap at scale, and *transparent in its instructions*: the rubric and prompts are explicit artifacts that can be inspected, criticized, versioned, and re-run identically. To put a number on “cheap,” the entire twelve-country, four-model sweep in this paper (on the order of 1,600 scoring and event calls) cost roughly \$15 in API charges and ran in a few hours on one laptop; an expert-panel coding of the same country-periods would cost orders of magnitude more in time and money. Its bias, however, is *opaque in origin*: it reflects the distribution of its training data [4, 9, 17], which over-represents English-language, recent, and Western sources, and it tends toward received consensus, likely flattening contested or under-documented episodes rather than flagging them.

It has no accountable author, and its scores can shift with model version or provider (Section 5.5).

Reliability as the bridge. The inter-coder logic that human panels use for humans applies to models: agreement across independent models is a measurable proxy for reliability. Where independent models agree on the moves, a score is more likely to reflect a shared signal than one model’s idiosyncrasy; where they disagree, the disagreement usefully localizes the contested cases a human panel would also flag. Reliability is necessary but not sufficient for validity, though: models trained on overlapping corpora can agree on a shared bias, which is why we treat both inter-model agreement and V-Dem consistency as evidence about *reliability and consensus*, not accuracy.

When to use which. We argue the LLM pipeline is best positioned as a *scalable first pass and a reproducible baseline* (generating candidate trajectories over many countries cheaply, surfacing where models disagree, and freeing expert attention for the contested cases) rather than as a replacement for expert coding where accountability and contested-case judgment are paramount.

7 Limitations

Several limitations bound what this proof of concept establishes. *Circularity of the consistency check.* V-Dem and similar indices are almost certainly in the models’ training data, so agreement with them cannot distinguish measuring the construct from recalling memorized values; we therefore frame the check as concurrent consistency, not accuracy, and disentangling the two (e.g., counterfactual-history probes, or ablating the events step) is future work. *Construct compression.* The two axes reduce vast, contested constructs to single numbers, and our single state-capacity indicator only partially aligns with the theory’s notion of state power. *Coverage and language bias.* Training corpora over-represent English-language, recent, and Western sources, weakest for exactly the cases we most need to read carefully (Iran, China, Chile, Colombia, and the Congo), and we prompt in English throughout. *Path dependence.* In-context anchoring can propagate an early mis-score through the whole trajectory; we do not ablate it here. *Model idiosyncrasy.* We report the mean-across-models atlas (Fig. 3) as the headline reading and quantify per-axis spread (Section 5.5); still, all four are English-centric LLMs and may share bias. *Scale and statistics.* Twelve countries and short per-country series preclude cross-national generalization or strong significance claims. We deliberately keep the study small; the natural next steps, a purpose-built human coding of

this exact task (beyond the V-Dem comparison we already report), prompt and period-granularity ablations, a non-English prompting probe, and more expert indices, are left to future work.

8 Conclusion

We presented, as a proof of concept, a method and a provider-agnostic tool that operationalize the Narrow Corridor’s (state, society) space into reproducible trajectories, together with a protocol for checking their consistency with expert consensus and their reliability across models, and released it as a one-command exercise. The resulting twelve-country atlas is a first quantified rendering of a framework that has been purely qualitative, and the accompanying consistency check and bias discussion frame honestly what such LLM-generated measurements are, and are not, good for.

Reproducibility

All code, prompts, the on-disk response cache, and every run (JSON, static plots, animations) are released at <https://github.com/esonghori/narrow-corridor-llm>. The exact experiments in this paper are reproduced by `paper/experiments/run_experiments.py` and `analysis.py`; see `paper/README.md`. The interactive gallery of animated trajectories is hosted at <https://esonghori.github.io/narrow-corridor-llm/>.

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A Example: one period's full prompts and responses

To make the two-call-per-period pipeline concrete, below is the *verbatim* exchange for a single period, the United Kingdom in 1829–1838, under `claude-opus-4-8`: the chain-of-thought events prompt and the model's reply, then the scoring prompt (which embeds the shared 0–10 rubric of Section 3 and the trajectory so far) and the schema-validated JSON score it returns.

```
=====
(1) EVENTS-AND-TRENDS PROMPT [chain-of-thought, step 1]
=====
Considering Daron Acemoglu and James A. Robinson's framework in *The Narrow Corridor: States, Societies, and the Fate of Liberty*, identify the major historical events and trends in United Kingdom during 1829-1838 that materially affected EITHER the power of the state OR the power of civil society.

Be concrete and specific to 1829-1838. Distinguish slow-moving trends (institutional, economic, demographic) from discrete events. For each item, note briefly whether it strengthened or weakened the state, society, or both.

Respond with two short bulleted lists:

Trends:
- ...

Events:
- ...

=====
(2) MODEL RESPONSE [events and trends]
=====
# UK, 1829-1838: State vs. Society in the Acemoglu-Robinson Framework

**Trends:**

- **Industrialization and urbanization**: The continued growth of manufacturing towns (Manchester, Birmingham, Leeds) concentrated new populations and created an assertive industrial middle class and urban working class---**strengthened society** by generating new mobilizing groups and demands for representation.

- **Growth of the popular press and associational life**: Expanding newspapers, political unions, and pamphleteering deepened the capacity for collective action and public opinion---**strengthened society**.

- **Fiscal-administrative modernization**: The gradual professionalization of governance (culminating in reforms like centralized poor-law administration) expanded the state's administrative reach and bureaucratic capacity---**strengthened the state**.

- **Rising economic distress among agricultural and industrial laborers**: Falling wages and mechanization fueled discontent, feeding movements like the emerging Chartist and trade-union agitation---**strengthened society's mobilization**, while pressuring the state.

- **Erosion of the old confessional/aristocratic settlement**: The long decline of exclusive Anglican-aristocratic monopoly on politics set the stage for enfranchisement of new groups---**shifted power toward society**.

**Events:**

- **Catholic Emancipation Act (1829)**: Removed most civil disabilities on Catholics, following mass mobilization (O'Connell's Catholic Association)---**strengthened society** (demonstrated concessions could be extracted) while showing the state adapting.

- **Metropolitan Police Act (1829)**: Peel's creation of the London police established a permanent, centralized coercive apparatus---**strengthened the state** (enhanced monopoly on order).

- **Swing Riots (1830-1831)**: Agrarian uprisings against mechanization and low wages; suppressed by state force but signaled rural society's capacity for disruption---**mobilized society**, prompted **state repression**.

- **Great Reform Act (1832)**: Redistributed seats to industrial towns and expanded the franchise to more middle-class men after intense extra-parliamentary pressure---**strengthened society** by broadening representation, though it excluded the working class.

- **Abolition of Slavery Act (1833)**: Product of sustained civil-society campaigning---**strengthened society's** demonstrated influence over policy.

- **Factory Act (1833)**: Regulated child labor and introduced inspectors---**strengthened both**: the state's regulatory reach and society's protective claims.

- **Poor Law Amendment Act (1834)**: Centralized and disciplined poor relief via workhouses and a national commission---**strengthened the state's** administrative power, provoking societal backlash.

- **Municipal Corporations Act (1835)**: Reformed and standardized local government, opening town councils to elected ratepayers---**strengthened both** local state capacity and civic participation.

- **Publication of the People's Charter (1838)**: Launched Chartism, articulating working-class demands (universal male suffrage, secret ballot)---**strengthened society's** organized mass mobilization.

*Caveat: Acemoglu and Robinson use this era as a case of Britain moving within "the corridor"--balanced growth of state and society through contested reform rather than either overwhelming the other.*
```


=====
(3) SCORING PROMPT [step 2; embeds the shared rubric + trajectory so far]
=====

You are placing the country of United Kingdom in the 2D space of Acemoglu and Robinson's 'The Narrow Corridor': the x-axis is the power of civil society and the y-axis is the power of the state. We are tracing the path from 1789 to 2020.

Use this fixed 0-10 rubric so values are comparable across periods and countries.

STATE POWER -- the capacity of the state to enforce its will, administer territory, collect taxes, and project authority:
0-2 collapsed / nonexistent: no effective central authority.
3-4 weak: limited reach, contested control, depends on local powerbrokers.
5-6 moderate: functioning bureaucracy and military with real but bounded reach.
7-8 strong: pervasive administration, effective coercion and revenue capacity.
9-10 overwhelming: near-total penetration of society; little it cannot enforce.

SOCIETY POWER -- the capacity of civil society to organize, mobilize, hold the state accountable, and resist domination (norms, associations, press, protest, independent institutions):
0-2 atomized: no autonomous organization; collective action near-impossible.
3-4 weak: sporadic, easily suppressed mobilization.
5-6 moderate: real associational life and periodic effective mobilization.
7-8 strong: dense civic institutions routinely constraining the state.
9-10 dominant: society pervasively shapes and checks state action.

Guidance for objectivity:
- Judge each period on its own terms; do not assume the country trends toward liberty, autocracy, or any particular corner of the space.
- Weigh evidence both for and against a shift before settling on a number.
- Avoid presentism: score by the standards and information of the period, not by how things turned out later.
- These two axes are independent: a strong state can coexist with a strong or a weak society. Score them separately.

Score the period 1829-1838, building on the established trajectory. Start from where United Kingdom stood at the end of 1819-1828, then apply the changes implied by the events and trends below. Report both the per-axis change and the resulting absolute values (anchored to the rubric, not just to the prior point). A period with little real change should show changes near 0.0.

Historical events and trends in 1829-1838:
UK, 1829-1838: State vs. Society in the Acemoglu-Robinson Framework

Trends:

- **Industrialization and urbanization**: The continued growth of manufacturing towns (Manchester, Birmingham, Leeds) concentrated new populations and created an assertive industrial middle class and urban working class---**strengthened society** by generating new mobilizing groups and demands for representation.
- **Growth of the popular press and associational life**: Expanding newspapers, political unions, and pamphleteering deepened the capacity for collective action and public opinion---**strengthened society**.
- **Fiscal-administrative modernization**: The gradual professionalization of governance (culminating in reforms like centralized poor-law administration) expanded the state's administrative reach and bureaucratic capacity---**strengthened the state**.
- **Rising economic distress among agricultural and industrial laborers**: Falling wages and mechanization fueled discontent, feeding movements like the emerging Chartist and trade-union agitation---**strengthened society's mobilization**, while pressuring the state.
- **Erosion of the old confessional/aristocratic settlement**: The long decline of exclusive Anglican-aristocratic monopoly on politics set the stage for enfranchisement of new groups---**shifted power toward society**.

Events:

- **Catholic Emancipation Act (1829)**: Removed most civil disabilities on Catholics, following mass mobilization (O'Connell's Catholic Association)---**strengthened society** (demonstrated concessions could be extracted) while showing the state adapting.
- **Metropolitan Police Act (1829)**: Peel's creation of the London police established a permanent, centralized coercive apparatus---**strengthened the state** (enhanced monopoly on order).
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- **Great Reform Act (1832)**: Redistributed seats to industrial towns and expanded the franchise to more middle-class men after intense extra-parliamentary pressure---**strengthened society** by broadening representation, though it excluded the working class.
- **Abolition of Slavery Act (1833)**: Product of sustained civil-society campaigning---**strengthened society's** demonstrated influence over policy.

- **Factory Act (1833)**: Regulated child labor and introduced inspectors--**strengthened both**: the state's regulatory reach and society's protective claims.
- **Poor Law Amendment Act (1834)**: Centralized and disciplined poor relief via workhouses and a national commission--**strengthened the state's** administrative power, provoking societal backlash.
- **Municipal Corporations Act (1835)**: Reformed and standardized local government, opening town councils to elected ratepayers--**strengthened both** local state capacity and civic participation.
- **Publication of the People's Charter (1838)**: Launched Chartism, articulating working-class demands (universal male suffrage, secret ballot)--**strengthened society's** organized mass mobilization.

Caveat: Acemoglu and Robinson use this era as a case of Britain moving within "the corridor"--balanced growth of state and society through contested reform rather than either overwhelming the other.

Trajectory so far (period: <society_power, state_power>):

1789-1798: <6.0, 6.5>
 1799-1808: <5.5, 7.5>
 1809-1818: <5.5, 7.5>
 1819-1828: <6.5, 7.5>

Return a JSON object with exactly these fields:

- society_power (float, 0-10): absolute society power at the end of 1829-1838
- state_power (float, 0-10): absolute state power at the end of 1829-1838
- society_change (float): society_power change vs. 1819-1828
- state_change (float): state_power change vs. 1819-1828
- key_event (string, <= 10 words: the defining event/trend driving the change)
- reasoning (string, 1-2 sentences justifying the values)

=====
 (4) MODEL RESPONSE [validated against the PeriodScore JSON schema]
 =====

```
{
  "society_power": 7.5,
  "state_power": 8.0,
  "society_change": 1.0,
  "state_change": 0.5,
  "key_event": "Great Reform Act (1832) and rise of mass mobilization",
  "reasoning": "Society surged as Catholic Emancipation, the Reform Act, abolition, and Chartism demonstrated dense associational life extracting concessions, while the state modernized via the Metropolitan Police, centralized Poor Law, and factory inspection\u2014balanced growth within the corridor."
}
```